

YM Gap Analysis — Two Distinct Concepts

A number-theoretic defect vs. the (unknown) Yang-Mills mass gap · no link claimed

1 · The Bridge Defect "Gap"

$gap_bridge = |q \cdot \kappa_{15}^m - P_5 - k \cdot \pi|$
 $= 0.0382906 < 1$ at $\kappa_{15} = 4.84330141945946$
 Status: VERIFIED ($gap_bridge < 1$). A Diophantine approximation defect.
 Units: dimensionless.
 Precision: ± 0.29 from κ_{15} uncertainty — only the verdict " $gap_bridge < 1$ " is robust.

2 · The Yang-Mills Mass Gap

gap_YM = lowest energy eigenvalue > 0 in $SU(3)$ pure gauge theory
Status: UNKNOWN. Not computed. Not claimed.
 Connection to gap_bridge : NONE PROVEN. This document makes no link.
 Required for any claim: a proof relating gap_bridge to gap_YM . Not provided.

3 · Open Questions

- 1. Is there a mathematical map from gap_bridge to gap_YM ? **OPEN**
- 2. Does κ_{15} have any physical meaning? **OPEN**
- 3. Are the P_k related to glueball states? **OPEN**

4 · What Exists In The Repo

- P_1 – P_5 : certified exceptional primes ($P_5 = 3,993,746,143,633$).
- P_6 – P_{20} : listed, BEYOND_TOLERANCE at κ_{15} .
- Wall 574 (Towers/YM/MassGap574.lean): the YM_mass_gap surface — OPEN, scalar-shadow only, NOT a brick. It carries NO numerical value and NO tie to P_5 .

Correction (honesty):
 The requested datum " $Brick_{574} = 3993746143627$ " does NOT exist in the repo — no such value appears anywhere, and it equals $P_5 - 6$. It was dropped as fabricated. Wall 574 is an index label, not a 13-digit integer near P_5 .

Verdict
 gap_bridge (number theory, verified < 1) and gap_YM (physics, unknown) are SEPARATE. No equality, no map, no claim.